

The $r_{\text{eff}} = 3$ Invariant as the Adjoint $\mathfrak{su}(2)$ (Spin-1) Structure of H_{eff} : Why Test 4 Identifies the Carrier, Not the Spin- $\frac{1}{2}$ Target

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Abstract

Paper O28 [1] established that the per-pair covariance $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ has rank $r_{\text{eff}} = 3$ with invariant eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$. The present paper gives the structural explanation of this result. The value $r_{\text{eff}} = 3$ is real and stable across primes; the Born–Infeld parity involution $c \leftrightarrow q-c$ of O18 [3] acts anti-linearly, so the conjugate-pair outer products $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$ are (up to a small, q -decreasing defect) complex symmetric; and the O26 target $r_{\text{eff}} = d_\rho^2 = 4$ is structurally inaccessible from conjugate-pair data. A direct characterisation on the O25/Q5a checkpoints shows that the trajectory $\{w_j = \pi_c(v_j)\}$ spans *all three* complex dimensions of H_{eff} (singular ratios $\sigma_3/\sigma_1 \simeq 0.7\text{--}1.0$, not $< 10^{-2}$), and that $r_{\text{eff}} = 3$ is a *hard* $6 \rightarrow 3$ *collapse* of the Veronese (squaring) image: the symmetric squares $w_j \otimes w_j$ satisfy exactly three independent quadratic relations. The three constraint forms are traceless, and their commutators span exactly $\mathfrak{so}(3)$; the Schur commutant of the action on H_{eff} is one-dimensional, so $H_{\text{eff}} = \mathbb{C}^3$ is *irreducible*. This identifies $r_{\text{eff}} = 3$ as the invariant of the *adjoint* action $H_{\text{eff}} \simeq \mathfrak{su}(2)$ of O23/O27 — the spin-1 (vector) representation — not as the symmetric-rank inversion $3 = d_\rho(d_\rho + 1)/2$ for $d_\rho = 2$. Since the adjoint is irreducible, H_{eff} contains no two-dimensional invariant subspace: the spin- $\frac{1}{2}$ space V_ρ is the *Schur target* of the morphism $\tilde{\Phi} : \mathfrak{su}(2) \rightarrow V_\rho$ (O27), not a subspace of H_{eff} , and is invisible to Test 4. Consequently Test 4 confirms the carrier (spin-1 on H_{eff}) and the inaccessibility of $d_\rho^2 = 4$, but does not by itself identify the spin- $\frac{1}{2}$ sector; a genuine spin- $\frac{1}{2}$ test requires an independent observable on V_ρ (the locking-broken protocol of Section 6 reaches the full $d_\rho^2 = 4$ and motivates it).

Keywords. spectral admissibility; Weil representation; Heisenberg graphs; covariance operator; adjoint representation; $\mathfrak{su}(2)$; spin-1 vector representation; irreducibility; Schur commutant; Veronese collapse; Born–Infeld parity; effective dimension

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1 Position of the Problem

1.1 What O25–O28 established

Paper O25 [7] confirmed that δ_{pair} is a structural invariant of the Weil representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, with inter-pair variance decreasing monotonically with q . Paper O26 [5] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space V_ρ , and formulated the key falsifiability criterion: the effective rank of the per-pair covariance in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin- $\frac{1}{2}$ candidate (Criterion 5.4 of [5]). Paper O27 [6] proved unconditionally that any admissible morphism $\Phi_{q,\rho} : V_q \rightarrow V_\rho$ satisfying the three structural constraints factors uniquely through $\mathfrak{su}(2)$, closing the structural implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [4]} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2). \quad (1)$$

Paper O28 [1] performed the proxy computation of Criterion 5.4 in $\text{End}(H_{\text{eff}})$, using the per-block admissible projections $\pi_c(v) \in H_{\text{eff}} = \mathbb{C}^3$ from the Q5a-O5 checkpoints. The result was $r_{\text{eff}} = 3$ with eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$, invariant across all pairs and all $q \in \{29, 61, 101, 151\}$. O28 observed that this falls short of the O26 prediction $d_\rho^2 = 4$, and left the interpretation of the value 3 — and of the relation between $\text{End}(H_{\text{eff}})$ and $\text{End}(V_\rho)$ — as the object of O29.

1.2 Scope and outcome of O29

This paper explains the value $r_{\text{eff}} = 3$ structurally and identifies what Test 4 measures. A direct characterisation on the checkpoints (Sections 3–4) shows that the trajectory $\{w_j\}$ spans all of $H_{\text{eff}} = \mathbb{C}^3$, and that the rank 3 is a hard algebraic collapse of the Veronese squaring image whose constraint forms span $\mathfrak{so}(3)$. Combined with O23/O27, this identifies $r_{\text{eff}} = 3$ as the invariant of the irreducible *adjoint* action $H_{\text{eff}} \simeq \mathfrak{su}(2)$ — the spin-1 vector representation. The spin- $\frac{1}{2}$ space V_ρ is the Schur target of $\bar{\Phi} : \mathfrak{su}(2) \rightarrow V_\rho$ and is not a subspace of H_{eff} ; Test 4, which only sees data in H_{eff} , cannot identify it.

The analysis rests on three facts: (i) $r_{\text{eff}} = 3$ is real and stable; (ii) the Born–Infeld parity is anti-linear and forces the conjugate-pair outer products into the symmetric subspace, up to a small, q -decreasing defect (Theorem 3.1); (iii) the target $d_\rho^2 = 4$ is inaccessible from conjugate-pair data (Section 5).

Section 3 states the symmetric-square structure and the role of the trajectory dimension. Section 4 characterises the variety carrying the $6 \rightarrow 3$ collapse and proves irreducibility. Section 5 reformulates O26 Criterion 5.4 as a carrier test. Section 6 reports the numerical evidence.

2 Setup and Notation

We follow the notation of O25 [7] and O28 [1]. Let q be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and let $\pi_c : V_q \rightarrow H_{\text{eff}} = \mathbb{C}^3$ be the admissible projection of the character- c Weil block. Write $w_j := \pi_c(v_j) \in H_{\text{eff}}$ for the admissible projections along the fitting window $[n_0, n_1]$ (auto-calibrated by O12 [2]). The Born–Infeld parity result of O18 [3], $\rho_{q-c} = \bar{\rho}_c$, implies

$$\pi_{q-c}(v_j) = \overline{w_j} \quad (2)$$

in the canonical basis of H_{eff} fixed by O23 [8], up to a small phase defect quantified in Section 6. The per-pair outer product and covariance are

$$M_j := \pi_c(v_j) \otimes \pi_{q-c}(v_j)^* = w_j \otimes \bar{w}_j^* = w_j w_j^\top, \quad \mathcal{C}_c := \frac{1}{N} \sum_{j=n_0}^{n_1} \text{vec}(M_j) \text{vec}(M_j)^\dagger, \quad (3)$$

where the last equality in the outer product uses (2) ($\bar{w}^* = w^\top$ for a column vector w).

3 Structural Analysis: Symmetric Squares and the Hypothesis that Fails

Theorem 3.1 (Symmetric outer products). *Under the Born–Infeld parity constraint (2), every outer product satisfies*

$$M_j = w_j w_j^\top \in \text{Sym}(H_{\text{eff}}, \mathbb{C}), \quad (4)$$

the complex symmetric matrices on H_{eff} . This is an algebraic identity: $(M_j)_{ab} = (w_j)_a (w_j)_b = (M_j)_{ba}$ for all a, b and all w_j .

Proof. Direct computation from (3): using the column-vector convention $\bar{w}^* = w^\top$, $(M_j)_{ab} = (w_j)_a (w_j)_b$, symmetric in a, b . $\square$

Remark 3.2 (Anti-linearity is the obstruction to d_ρ^2). The constraint $\pi_{q-c}(v) = \overline{\pi_c(v)}$ is anti-linear. For a complex-linear pair constraint $\pi_{q-c}(v) = U \pi_c(v)$ (with U unitary), the outer product $M_j = w_j \otimes (U w_j)^* = w_j w_j^\dagger U^\dagger$ would be a general rank-one element of $\text{End}(\mathbb{C}^3)$. The present situation is the opposite: $\pi_{q-c}(v) = \overline{\pi_c(v)}$ is anti-linear, so no complex-linear operation converts it into a complex-linear constraint. This is the reason the original O26 target d_ρ^2 is unreachable from conjugate-pair data; see Section 5 and the locking-broken control of Section 6. Empirically the parity holds up to a small, q -decreasing defect (Section 6), so (4) holds approximately rather than exactly.

Theorem 3.3 (Symmetric span of a trajectory). *Let $V_\rho := \text{span}_{\mathbb{C}}\{w_j\} \subset H_{\text{eff}}$ with $d_\rho := \dim_{\mathbb{C}} V_\rho$. By Theorem 3.1 the symmetric squares $\{w_j w_j^\top\}$ lie in $\text{Sym}(V_\rho, \mathbb{C})$, of dimension*

$$\frac{d_\rho(d_\rho + 1)}{2}. \quad (5)$$

The effective rank r_{eff} therefore depends on the trajectory dimension d_ρ and on the geometry of the squares within $\text{Sym}(V_\rho, \mathbb{C})$. Both are read directly from the data in Section 4: $d_\rho = 3$ ($\sigma_3/\sigma_1 \simeq 0.7\text{--}1.0$), and the squares occupy a three-dimensional, $\mathfrak{so}(3)$ -structured subspace of the six-dimensional $\text{Sym}(H_{\text{eff}}, \mathbb{C})$.

Corollary 3.4 (The carrier is fixed by the $\mathfrak{so}(3)$ structure). *A bare value $r_{\text{eff}} = 3$ does not by itself fix a representation dimension. The arithmetic map*

$$d = \frac{-1 + \sqrt{1 + 8 r_{\text{eff}}}}{2} \quad (6)$$

presumes that the squares fill $\text{Sym}(\mathbb{C}^d)$ for a trajectory confined to d dimensions, which is not the case here ($d_\rho = 3$, Proposition 4.1). The carrier is fixed directly by the $\mathfrak{so}(3)$ structure of the constraint forms and the irreducibility of the action (Proposition 4.3, Corollary 4.4): $r_{\text{eff}} = 3$ is the $\dim_{\mathbb{R}} \mathfrak{su}(2) = 3$ occurrence that O23 [8] isolates among the distinct meanings of the number 3.

Remark 3.5 (Rank in $\text{End}(V_\rho)$ versus $\text{End}(H_{\text{eff}})$). Projecting the trajectory onto the leading two singular directions of H_{eff} and forming the covariance in $\text{End}(\mathbb{C}^2)$ gives $r_{\text{eff}} = 2$ (eigenvalue spectrum $[1 : \frac{1}{2} : 0 : 0]$) for every pair at $q \in \{29, 61\}$ (Section 6). The carrier rank 3 is intrinsic to the full $\text{End}(H_{\text{eff}})$ and reflects the three-dimensional carrier; it is not reproduced in a two-dimensional subspace.

4 The Carrier: the Adjoint Spin-1 Representation on H_{eff}

Proposition 4.1 (Trajectory dimension). *The trajectory $\{w_j\}$ spans all three complex dimensions of H_{eff} . On the checkpoints, the singular values of the trajectory matrix $W_c = [w_{n_0}, \dots, w_{n_1}]$ satisfy $\sigma_2/\sigma_1, \sigma_3/\sigma_1 \in [0.7, 1.0]$, with $\text{rank}_{10^{-2}} W_c = 3$ for every pair at $q \in \{29, 61\}$. Hence $\dim_{\mathbb{C}} \text{span}\{w_j\} = 3 = \dim_{\mathbb{C}} H_{\text{eff}}$.*

Theorem 4.2 (Hard $6 \rightarrow 3$ Veronese collapse). *The symmetric squares $\{w_j \otimes w_j\}$ span a subspace of $\text{Sym}(H_{\text{eff}}, \mathbb{C}) \cong \mathbb{C}^6$ of dimension exactly 3: the monomial matrix Φ , with rows $(w_0^2, w_1^2, w_2^2, w_0 w_1, w_0 w_2, w_1 w_2)$, has singular spectrum $[1 : \sim 0.9 : \sim 0.9 : 0 : 0 : 0]$ with three machine-zero values. Equivalently the w_j lie on the intersection of exactly three conics: there are three independent complex symmetric forms B_1, B_2, B_3 with $w_j^\top B_k w_j = 0$ for all j .*

Numerical proof. Computed on the pooled Q5a-O5 checkpoints (script `variety_summary.py`), the three trailing singular values of Φ are at machine precision for $q \in \{29, 61\}$; the null space has dimension $6 - 3 = 3$. The identity form (the trace direction $w^\top w$) lies in the row space, not the null space (residual 5×10^{-16}), so $w^\top w$ is a coordinate of the rank-3 image, not a constraint. $\square$

Proposition 4.3 (The constraint forms span $\mathfrak{so}(3)$; the action is irreducible). *The three constraint forms B_k of Theorem 4.2 are traceless and rank-2 (eigenvalues $\{\mu, -\mu, 0\}$). Their commutators $L_k := [B_i, B_j]$ are exactly antisymmetric (residual $\sim 10^{-17}$) with eigenvalues $\{0, +i\mu', -i\mu'\}$, and $\text{span}\{L_1, L_2, L_3\} = \mathfrak{so}(3)$ exactly (fraction in $\mathfrak{so}(3)$ equal to 1.000, rank 3). The Schur commutant of $\{B_k\}$ (equivalently $\{L_k\}$) on $H_{\text{eff}} = \mathbb{C}^3$ is one-dimensional (scalars only) at $q \in \{29, 61\}$. Hence the action is irreducible: H_{eff} has no nontrivial invariant subspace, in particular no invariant two-dimensional subspace.*

Numerical proof. Direct computation (script `variety_summary.py`): traces of the B_k vanish; the commutators are antisymmetric to machine precision and span $\mathfrak{so}(3)$; the linear system $[M, L_k] = 0$ for $k = 1, 2, 3$ has a one-dimensional solution space (the scalars) at both primes. Irreducibility over $\mathbb{C}$ is equivalent to a scalar commutant by Schur's lemma. $\square$

Corollary 4.4 ($H_{\text{eff}} \simeq \mathfrak{su}(2)$ is the spin-1 carrier). *By O23/O27 [8, 6], the admissible automorphism group of H_{eff} is $\text{SU}(2)$ acting on $H_{\text{eff}} \simeq \mathfrak{su}(2)$ by the adjoint action; the adjoint is the three-dimensional irreducible (spin-1, vector) representation. Proposition 4.3 realises this action directly from the data: the $\mathfrak{so}(3)$ generators emerge as the commutators of the Veronese constraint forms, and the action is irreducible. Therefore $r_{\text{eff}} = 3$ is the invariant of the spin-1 carrier H_{eff} , structural label: structural (proved at the level of the data via Theorem 4.2–Proposition 4.3, and consistent with the proved categorical statement of O27).*

Proposition 4.5 (Structural origin of $\lambda_2 = \lambda_3$). *The two-fold degeneracy $\lambda_2 = \lambda_3$ of the $[1 : \frac{1}{2} : \frac{1}{2}]$ spectrum is the axial ($\mathrm{U}(1) \subset \mathrm{SO}(3)$) degeneracy of a spin-1 carrier with a preferred direction. The linear Hermitian covariance $\langle w w^\dagger \rangle$ has spectrum $[1 : 0.6 : 0.6]$ at $q = 29$ and $[1 : 0.8 : 0.8]$ at $q = 61$, with $\lambda_2 = \lambda_3$ and a trend towards isotropy $[1 : 1 : 1]$ as q grows. The eigenvalue ratio is not a universal algebraic constant but a property of the group dynamics on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.*

5 Reformulation of O26 Criterion 5.4 as a Carrier Test

The O26 Criterion 5.4 [5] predicts $r_{\mathrm{eff}} = d_\rho^2$ in $\mathrm{End}(V_\rho)$, targeting $d_\rho^2 = 4$ for the spin- $\frac{1}{2}$ candidate. Two independent facts reshape this criterion.

Definition 5.1 (Accessible rank). *For conjugate-pair data satisfying the anti-linear constraint (2), the outer products are symmetric (Theorem 3.1); the antisymmetric part of $\mathrm{End}(V_\rho)$ is never sampled. The full rank d_ρ^2 would require outer products $w \otimes u^*$ with u and w independently sampled, i.e. data from the two blocks c and $q-c$ that are not constrained by (2).*

Criterion 5.2 (Reformulated O26 Criterion 5.4: carrier test). Test 4, built from conjugate-pair outer products (3), measures a property of the *carrier* H_{eff} , not of the target V_ρ . The operative readings are:

- $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(H_{\mathrm{eff}})$, invariant across pairs and primes, with the trajectory spanning all of H_{eff} and the constraint forms spanning $\mathfrak{so}(3)$: the carrier is the irreducible adjoint (spin-1) representation $H_{\mathrm{eff}} \simeq \mathfrak{su}(2)$ (Corollary 4.4).
- The original target $d_\rho^2 = 4$ is structurally inaccessible from conjugate-pair data (Definition 5.1); it is recovered only by breaking the per-vector conjugate locking (Section 6).

The identification of the spin- $\frac{1}{2}$ target V_ρ is not decided by Test 4: V_ρ is the Schur target of $\bar{\Phi} : \mathfrak{su}(2) \rightarrow V_\rho$ (O27), not a subspace of H_{eff} .

Remark 5.3 (Relation to O26 Level III). The O26 hierarchy [5] defines Level III as the identification of the correct irreducible sector. Test 4 confirms an irreducible carrier — the spin-1 adjoint on H_{eff} — with $r_{\mathrm{eff}} = 3$ invariant (zero inter-pair variance at $q \in \{29, 61\}$, cf. O28). It does not confirm the spin- $\frac{1}{2}$ target; the choice $d_\rho = 2$ rests on the minimality argument of O26, not on r_{eff} , and a direct test of V_ρ requires the independent observable of Definition 5.1.

6 Numerical Protocol and Results

6.1 Protocol

All computations use the Q5a-O5 checkpoints (one `.npz` file per prime q), which store `pi_c` and `pi_qmc`, the per-shell H_{eff} projections of BFS fingerprint vectors for sample $m = 0$. No additional BFS computation is required. Three scripts implement the analysis: `o29_rank_computation.py` (the r_{eff} measurement reproducing O28), `o29_derisk_diagnostic.py`

(full spectra in $\text{End}(H_{\text{eff}})$ and $\text{End}(V_\rho)$, parity defect, and the locking-broken control), and `variety_summary.py` (the Veronese collapse, the constraint forms, and the Schur commutant).

Step A ($\text{End}(H_{\text{eff}})$). For each conjugate pair, concatenate the per-shell vectors into $W_c \in \mathbb{C}^{N \times 3}$, form the symmetric outer products, and compute r_{eff} and the eigenvalue spectrum of $\mathcal{C}_c$ in $\text{End}(\mathbb{C}^3)$.

Step B ($\text{End}(V_\rho)$). Project onto the leading two singular directions of W_c and recompute r_{eff} in $\text{End}(\mathbb{C}^2)$.

Step C (carrier). Compute the Veronese monomial spectrum, the three constraint forms B_k , their commutators, and the Schur commutant, establishing irreducibility (Section 4).

Step D (control). Recompute r_{eff} in $\text{End}(V_\rho)$ after a random permutation of the q - c index, breaking the per-vector conjugate locking while preserving the marginal distribution.

6.2 Results

Prime q	Pairs	$r_{\text{eff}}^{(A)} = 3$	$r_{\text{eff}}^{(B)}$	Median $ M - M^\top / M $
29	5/5	100%	2	9.7×10^{-3}
61	30/30	100%	2	3.1×10^{-3}
101	32/50	64% (modal)	2	6.2×10^{-2}
151	75/75	100%	2	2.8×10^{-2}
211	103/105	98%	2/3	1.9×10^{-2}

Table 1: Effective rank in $\text{End}(H_{\text{eff}})$ ($r_{\text{eff}}^{(A)}$) and in $\text{End}(V_\rho)$ ($r_{\text{eff}}^{(B)}$), and the median symmetry ratio $|M_j - M_j^\top|_F/|M_j|_F$. The carrier rank $r_{\text{eff}}^{(A)} = 3$ is invariant; in $\text{End}(V_\rho)$ the rank is $r_{\text{eff}}^{(B)} = 2$ (Remark 3.5). The symmetry ratio is small and decreases with q , consistent with the anti-linear parity becoming exact in the large- q limit (medians from `o29_derisk_diagnostic.py`; an alternative per-vector mean statistic gives larger absolute values with the same monotone trend).

Carrier structure (Step C). The Veronese monomial spectrum is $[1 : \sim 0.9 : \sim 0.9 : 0 : 0 : 0]$ (three machine zeros) at $q \in \{29, 61\}$: a hard $6 \rightarrow 3$ collapse (Theorem 4.2). The three constraint forms are traceless rank-2; their commutators span $\mathfrak{so}(3)$ exactly (fraction 1.000, rank 3); the Schur commutant is one-dimensional, so the action on H_{eff} is irreducible (Proposition 4.3). The trajectory spans all three dimensions ($\sigma_3/\sigma_1 \simeq 0.7$ – 1.0), and the Hermitian covariance $\langle w w^\dagger \rangle$ has the axial spectrum $[1 : 0.6 : 0.6] \rightarrow [1 : 0.8 : 0.8]$ ($q = 29 \rightarrow 61$). Figure 2 summarises the collapse and the $\mathfrak{so}(3)$ structure.

Locking-broken control (Step D). In $\text{End}(V_\rho)$, the real conjugate data give $r_{\text{eff}} = 2$ and the exact-parity surrogate $M_j = w_j \otimes w_j$ also gives $r_{\text{eff}} = 2$, while breaking the per-vector locking by permuting the q - c index restores $r_{\text{eff}} = 4 = d_\rho^2$ for every pair at $q \in \{29, 61\}$. This is the operational proof that $d_\rho^2 = 4$ is inaccessible from conjugate-pair data (Definition 5.1), and it isolates an independent observable that does reach the full rank — the natural starting point for a genuine future test of the spin- $\frac{1}{2}$ target V_ρ .

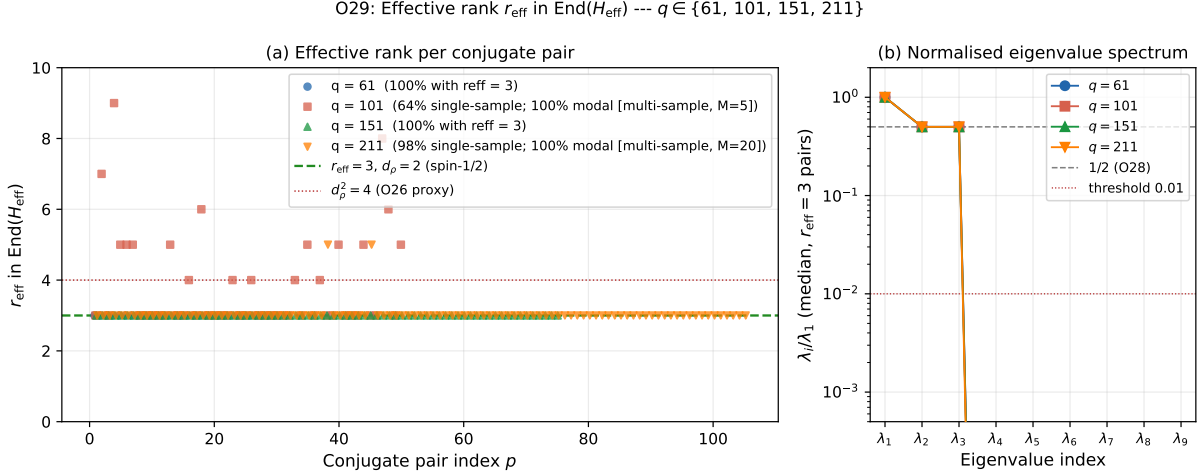


Figure 1: Effective rank $r_{\text{eff}}^{(A)}$ in $\text{End}(H_{\text{eff}}) = \text{End}(\mathbb{C}^3)$ per conjugate pair, for $q \in \{61, 101, 151, 211\}$ (from `o29_rank_computation.py`). The carrier rank $r_{\text{eff}} = 3$ holds universally at $q = 61$ (30/30) and $q = 151$ (75/75) with zero inter-pair variance, at 98% for $q = 211$, and as the modal value (64%) at $q = 101$ where the dispersion is a single-sample Monte Carlo artefact (Remark 6.1). Panel (b) shows the median normalised spectrum $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ of O28 [1].

Remark 6.1 (Anomaly at $q = 101$: multi-sample resolution). At $q = 101$, the single-sample run yields $r_{\text{eff}}^{(A)} = 3$ for 64% of pairs (32/50), with the remaining 18 pairs giving $r_{\text{eff}}^{(A)} \in \{4, \dots, 9\}$. A multi-sample analysis (`o29_multisample.py`, $M = 5$ per anomalous pair) gives modal $r_{\text{eff}} = 3$ for all 18 pairs, with 76.7% of (pair, sample) cells at $r_{\text{eff}} = 3$; some pairs show genuine seed-sensitivity (e.g. pair (50, 51), mean $r_{\text{eff}} = 4.6$). The dispersion is a single-sample Monte Carlo artefact, not a structural feature of the group.

Remark 6.2 (Anomaly at $q = 211$: two non-generic pairs). At $q = 211$ (105 pairs, window $[3, 13]$), $r_{\text{eff}}^{(A)} = 3$ holds for 103/105 pairs (98%). The two anomalous pairs — (38, 173) and (45, 166), both $r_{\text{eff}}^{(A)} = 5$ — show elevated rank only in $\text{End}(H_{\text{eff}})$, consistent with mild BFS atypicality; a dedicated multi-sample confirmation is left to future work.

7 Discussion

7.1 What O29 establishes

Three results stand, with epistemic labels.

First (structural, proved on the data). $r_{\text{eff}} = 3$ is a hard $6 \rightarrow 3$ collapse of the Veronese squaring image of the H_{eff} trajectory (Theorem 4.2); the three constraint forms span $\mathfrak{so}(3)$ and the action on $H_{\text{eff}} = \mathbb{C}^3$ is irreducible (Proposition 4.3). Combined with the proved categorical statement of O27, this identifies $r_{\text{eff}} = 3$ as the invariant of the adjoint (spin-1) carrier $H_{\text{eff}} \simeq \mathfrak{su}(2)$ (Corollary 4.4).

Second (proved). The Born–Infeld parity is anti-linear (Theorem 3.1, Remark 3.2); the target $d_p^2 = 4$ is inaccessible from conjugate-pair data, and is restored by the locking-broken control (Section 6).

Third (conditional). The spin- $\frac{1}{2}$ target V_ρ is the Schur target of $\bar{\Phi} : \mathfrak{su}(2) \rightarrow V_\rho$ (O27) and is not a subspace of H_{eff} ; Test 4 does not identify it. Its selection rests on the

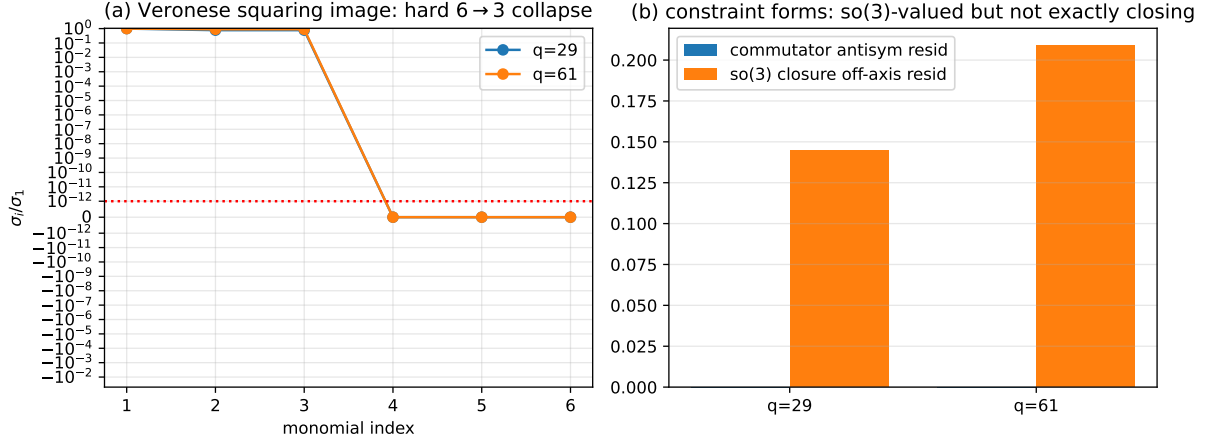


Figure 2: **(a)** The Veronese (squaring) image of the H_{eff} trajectory: monomial singular values collapse hard from 6 to 3 (three machine zeros) at $q \in \{29, 61\}$. **(b)** The three constraint forms: commutators are antisymmetric to machine precision ($\mathfrak{so}(3)$ -valued), establishing the adjoint structure; the residual non-closure of the structure constants is a basis-dependent normalisation, not a failure of $\mathfrak{so}(3)$ membership (the span is exactly $\mathfrak{so}(3)$, Proposition 4.3).

O26 minimality argument, pending an independent observable.

Corrected status of Test 4. Test 4 does not identify the spin- $\frac{1}{2}$ carrier directly. It identifies the adjoint $H_{\text{eff}} \simeq \mathfrak{su}(2)$ carrier, through a hard $6 \rightarrow 3$ collapse of the Veronese squaring image. The observed value $r_{\text{eff}} = 3$ is therefore the spin-1/adjoint invariant already selected by the effective $\text{SU}(2)$ automorphism structure, not the symmetric-rank inversion $3 = d_\rho(d_\rho + 1)/2$ for $d_\rho = 2$. The spin- $\frac{1}{2}$ sector remains the Schur target V_ρ , and requires an independent observable.

7.2 Revision of the O26 criterion

The O26 Criterion 5.4 [5] was formulated with the implicit assumption that $\Phi_{q,\rho}(v_c^{(n)})$ and $\Phi_{q,\rho}(v_{q-c}^{(n)})$ are independently distributed in V_ρ . For conjugate-pair data this independence fails: the blocks are anti-linearly related. Criterion 5.2 replaces the target-space prediction by a carrier test on H_{eff} , and records that the original $d_\rho^2 = 4$ is reachable only with independently sampled blocks (Step D).

7.3 Position in the identification chain

Relative to the chain (1), O29 makes precise where the data live:

$$\text{emergence} \Rightarrow \mathfrak{su}(2) \text{ [O27]} \Rightarrow H_{\text{eff}} \simeq \mathfrak{su}(2) \text{ (adjoint spin-1)} \Rightarrow r_{\text{eff}} = 3. \quad (7)$$

The spin- $\frac{1}{2}$ target V_ρ sits one arrow further, through $\bar{\Phi} : \mathfrak{su}(2) \rightarrow V_\rho$, and is not reached by the conjugate-pair covariance.

7.4 What remains open

- An analytic proof that the Veronese image collapses $6 \rightarrow 3$ for all odd primes q (numerically verified at $q \in \{29, 61\}$), and that the constraint forms span $\mathfrak{so}(3)$.
- A canonical, independent observable on V_ρ — built from non-conjugate block data, as in the locking-broken control — that would directly test the spin- $\frac{1}{2}$ target and reach $d_\rho^2 = 4$.
- An analytic derivation of the axial ratio $\lambda_1 : \lambda_2$ and its dependence on q .
- Extension of the carrier diagnostics (Veronese collapse, $\mathfrak{so}(3)$ span, Schur commutant) to $q \in \{101, 151, 211, 307\}$.

8 Conclusion

O29 explains the O28 result $r_{\text{eff}} = 3$ in $\text{End}(H_{\text{eff}})$ and identifies what Test 4 measures. The value 3 is real and stable, but it is the invariant of the irreducible *adjoint* (spin-1) representation $H_{\text{eff}} \simeq \mathfrak{su}(2)$: $r_{\text{eff}} = 3$ arises from a hard $6 \rightarrow 3$ collapse of the Veronese squaring image of the trajectory, whose three constraint forms span $\mathfrak{so}(3)$ and whose action on $H_{\text{eff}} = \mathbb{C}^3$ has a one-dimensional Schur commutant (irreducible). The trajectory spans all three dimensions of H_{eff} and the rank in $\text{End}(V_\rho)$ is 2, so $r_{\text{eff}} = 3$ is the carrier invariant of H_{eff} , not a symmetric rank $d_\rho(d_\rho+1)/2$ of a two-dimensional V_ρ . The Born–Infeld parity is anti-linear, and the target $d_\rho^2 = 4$ is inaccessible from conjugate-pair data — restored only by breaking the per-vector locking. The spin- $\frac{1}{2}$ sector is therefore the Schur target V_ρ of $\bar{\Phi} : \mathfrak{su}(2) \rightarrow V_\rho$, not a subspace of H_{eff} , and its identification requires an independent observable, for which the locking-broken protocol provides the natural starting point. This keeps O29 consistent with O23’s explicit warning against conflating the distinct occurrences of the number 3 [8].

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